

Technical Specifications

Version: 1.0

Purpose

This document defines the technical specifications of Trading Platform Pro. It complements the Product Vision, Project Overview and Software Architecture Document by specifying implementation constraints, technologies and technical requirements.

Technology Stack

- Python 3.13
- PySide6
- SQLite
- SQLAlchemy
- Dependency Injector
- Pytest
- Ruff
- GitHub Actions

Target Platforms

- Windows (Primary)
- Linux (Future)
- macOS (Future)

Application Architecture

- Clean Architecture
- Domain-Driven Design
- Event-Driven Architecture
- Modular Monolith

Core Technologies

- Dependency Injection
- AsyncIO
- Structured Logging
- YAML Configuration
- Event Bus

Data Storage

- SQLite
- Repository Pattern
- Configuration Files (YAML)

External Integrations

- Interactive Brokers
- Market Data Providers
- Plugin Interfaces

Performance Requirements

- Fast startup
- Responsive UI
- Deterministic execution
- Low memory footprint

Security Requirements

- No secrets in source control
- Externalized configuration
- Secure credential storage
- Structured logging

Quality Requirements

- Full type hints
- Ruff compliant
- Automated testing
- Documentation-first

Related Documents

- [Project_Overview.md](#)
- [Product_Vision.md](#)
- [Architecture.md](#)
- [Coding_Standards.md](#)
- [Development_Guidelines.md](#)